

PAYERNE, Switzerland

WMO: 066100

Lat: 46.8117N

Lon: 6.9425E

Elev: 1609

StdP: 13.86

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	21.0	12.2	10.8	20.0	16.1	13.1	23.8	22.1	46.5	19.8	44.1	4.9	50	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	86.6	67.7	83.1	66.5	79.9	65.4	69.2	82.4	67.8	79.9	66.4	77.4	5.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	97.6	74.1	63.6	93.3	72.6	62.4	89.4	71.1	34.2	82.8	33.0	80.1	31.9	77.6	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.1	14.5	12.4	12.1	91.4	4.8	3.8	8.7	94.1	5.9	96.3	3.2	98.5	-0.2	101.2		
			11.4	72.1	4.8	1.4	7.9	73.2	5.1	74.0	2.4	74.8	-1.1	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	33.6	35.7	42.8	49.1	56.7	63.6	67.2	66.3	58.6	51.0	41.0	34.9
	DBStd	13.50	7.19	7.24	6.64	6.31	5.95	6.40	5.61	5.57	5.74	6.19	6.43	6.89
	HDD50	2063	508	401	236	93	12	1	0	5	63	276	468	
	HDD65	5706	972	820	689	478	262	102	42	52	203	434	720	932
	CDD50	2108	0	1	12	65	221	409	532	505	263	94	6	0
	CDD65	278	0	0	0	0	6	60	109	92	11	0	0	0
	CDH74	2490	0	0	0	5	89	551	984	776	84	1	0	0
	CDH80	689	0	0	0	0	10	139	301	233	6	0	0	0
Wind	WSAvg	4.4	4.9	5.2	5.5	4.7	4.5	4.2	4.1	3.6	3.7	3.5	4.2	4.9
Precipitation	PrecAvg	34.2	2.3	2.0	2.1	2.5	3.3	3.7	3.6	3.6	2.9	2.7	3.1	2.6
	PrecMax	50.5	5.9	5.9	5.0	6.3	6.1	7.4	9.3	11.2	7.0	5.5	7.2	6.9
	PrecMin	22.9	0.2	0.0	0.2	0.5	1.3	0.9	0.8	0.6	0.4	0.2	0.2	0.0
	PrecStd	6.5	1.4	1.3	1.2	1.5	1.2	1.8	1.9	2.0	1.9	1.5	1.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	56.7	65.8	74.1	81.1	88.8	91.7	91.0	80.7	71.3	60.7	53.4
		MCWB	47.5	47.8	51.7	58.2	63.5	69.1	68.0	67.9	64.6	59.2	53.1	47.4
	2%	DB	50.1	52.2	61.5	69.5	76.4	84.0	87.0	85.8	76.5	66.5	55.8	50.3
		MCWB	45.7	45.3	50.0	55.3	61.2	66.9	67.4	67.4	63.9	58.2	50.7	46.2
	5%	DB	46.8	49.0	57.5	65.6	72.8	80.3	83.1	82.1	72.6	63.1	52.6	47.6
		MCWB	43.1	43.6	48.4	53.4	59.3	65.9	66.5	66.6	62.0	56.8	48.8	44.1
	10%	DB	44.0	46.0	54.0	61.7	68.9	76.6	79.6	78.3	69.0	60.2	49.9	45.1
		MCWB	40.9	41.8	46.6	51.5	57.9	64.1	65.1	65.3	60.6	55.6	46.9	42.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	48.4	53.4	59.1	66.0	71.6	71.0	70.8	67.2	61.5	53.9	48.9
		MCDB	52.6	54.5	63.1	71.6	78.0	85.2	85.2	84.0	77.0	67.5	58.9	51.8
	2%	WB	46.0	45.9	51.0	56.2	63.0	68.7	69.0	69.0	65.0	59.5	51.5	46.7
		MCDB	49.5	51.2	59.3	67.3	73.3	80.6	82.8	82.0	73.7	64.4	55.0	49.6
	5%	WB	43.4	44.0	49.1	54.1	60.7	66.9	67.5	67.5	63.1	57.8	49.2	44.6
		MCDB	46.7	48.6	56.3	63.9	70.0	77.6	80.2	79.2	70.6	62.0	52.0	47.2
	10%	WB	41.0	41.9	47.2	52.2	58.6	65.0	66.0	65.9	61.3	55.9	47.3	42.3
		MCDB	43.6	45.7	53.4	60.5	67.1	74.8	77.3	76.2	67.5	59.6	49.5	44.9
Mean Daily Temperature Range	5% DB	MDBR	8.7	12.1	16.3	18.3	18.2	19.1	20.2	19.2	17.2	13.8	9.9	8.5
		MCDBR	12.5	17.4	23.1	26.4	25.4	25.3	26.8	26.2	22.5	18.3	14.1	11.6
		MCWBR	9.0	11.8	13.3	13.9	12.4	11.3	10.5	10.5	11.3	10.9	9.8	8.5
	5% WB	MCDBR	11.8	16.1	20.8	24.2	22.7	23.4	24.1	23.3	20.5	16.6	13.1	10.5
	MCWBR	8.9	11.4	12.5	13.1	11.6	11.1	10.1	10.1	10.8	10.4	9.6	8.1	
Clear-Sky Solar Irradiance	taub	0.289	0.315	0.353	0.392	0.403	0.404	0.403	0.394	0.373	0.351	0.317	0.288	
	taud	2.426	2.365	2.329	2.260	2.277	2.316	2.312	2.349	2.390	2.438	2.472	2.446	
	Ebn at Noon	256	271	275	273	274	273	272	270	266	254	243	243	
	Edh at Noon	22	28	34	40	41	40	39	36	32	26	21	19	
All-Sky Solar Radiation	RadAvg	408	679	1046	1384	1587	1881	1836	1598	1242	797	440	331	
	RadStd	49	100	153	200	143	176	160	147	111	80	58	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	+1.57	+0.67	+0.70	N/A	N/A	N/A	N/A
(47)	Regional (107 neighbors)	N/A	N/A	N/A	+1.33	+0.72	+0.54	N/A	N/A	+103	+32

Nomenclature: See separate page